

Script Arguments in Bash (Terminal Style)

Definition:

Script arguments are values passed to a Bash script when it is executed. They make scripts more dynamic and reusable.

Accessing Script Arguments

Special variables let you access command-line arguments easily.

Common Variables:

`$0` -> Script name
`$1` -> First argument
`$2` -> Second argument
`$#` -> Number of arguments
`$@` -> All arguments separately
`$*` -> All arguments as one string
`$?` -> Exit status of last command
`$$` -> PID of current script

Example 1: Simple Argument Access

```
#!/bin/bash
echo "Script name: $0"
echo "First argument: $1"
echo "Second argument: $2"
echo "Total arguments: $#"
```

Run:

```
./script.sh apple banana
```

Output:

```
Script name: ./script.sh
```

First argument: apple
Second argument: banana
Total arguments: 2

Example 2: Loop Through Arguments

```
#!/bin/bash
echo "You passed $# arguments."
for arg in "$@"
do
    echo "Argument: $arg"
done
```

Output:

```
You passed 3 arguments.
Argument: 10
Argument: 20
Argument: 30
```

Example 3: Using Arguments in Calculations

```
#!/bin/bash
sum=$(( $1 + $2 ))
echo "Sum: $sum"
```

Run:

```
./script.sh 5 7
```

Output:

```
Sum: 12
```

Example 4: Handling Missing Arguments

```
#!/bin/bash
```

```
if [ $# -lt 2 ]; then
    echo "Usage: $0 num1 num2"
    exit 1
fi
echo "You entered $1 and $2"
```

Output:

```
Usage: ./script.sh num1 num2
```

Example 5: Using shift Command

```
#!/bin/bash
echo "All arguments: $@"
shift
echo "After shift: $@"
```

Run:

```
./script.sh a b c
```

Output:

```
All arguments: a b c
After shift: b c
```

Summary Table

Variable	Description	Example
\$0	Script name	./myscript.sh
\$1, \$2...	Positional arguments	\$1 -> first arg
\$#	Number of arguments	3
\$@	All arguments (separately)	arg1 arg2 arg3
\$*	All arguments (single str)	"arg1 arg2 arg3"
\$\$	PID of script	12345

\$?	Exit status	0 (success)	
\$!	PID of last bg process	-	